



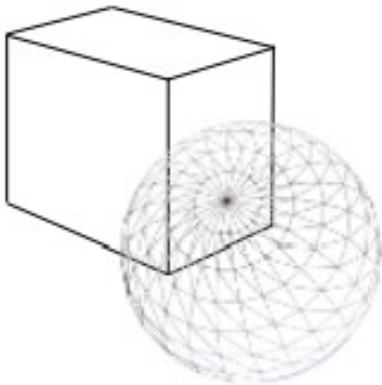
Illumination

Instructor: Dinesh Maharjan

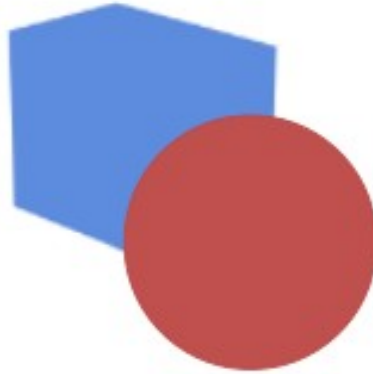
2022

Illumination

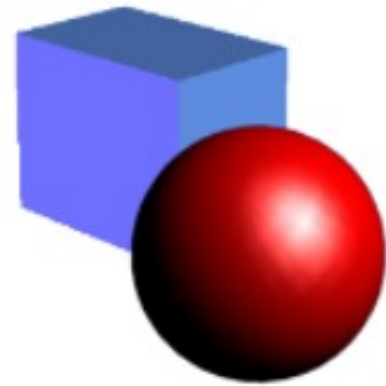
- Third stage of graphics pipeline deals with assigning color to object scene.
- Color helps to perceive depth.
- Color provides impression of 3D



wireframe



hidden surfaces



Next step...



Illumination

- Light on a surface is
- Absorbed
- Reflected and
- Transmitted.
- The amount of reflected light determines the color and brightness of the object.
-



Sources of Light

- Light source can be
- Point source
- Infinitely distance source
- Directional source
- Ambient Source
- We will consider point source and ambient source.



Light Sources

- Point source emits light in all direction equally.
- Is characterized by position and density of emitted light
- Infinitely distant source (sun) can be modeled with point source
- Such source is characterized only by intensity emitted.
- Directional source is a point source that emits light within angular limit.
- Characterized by source position, intensity and angular limit.

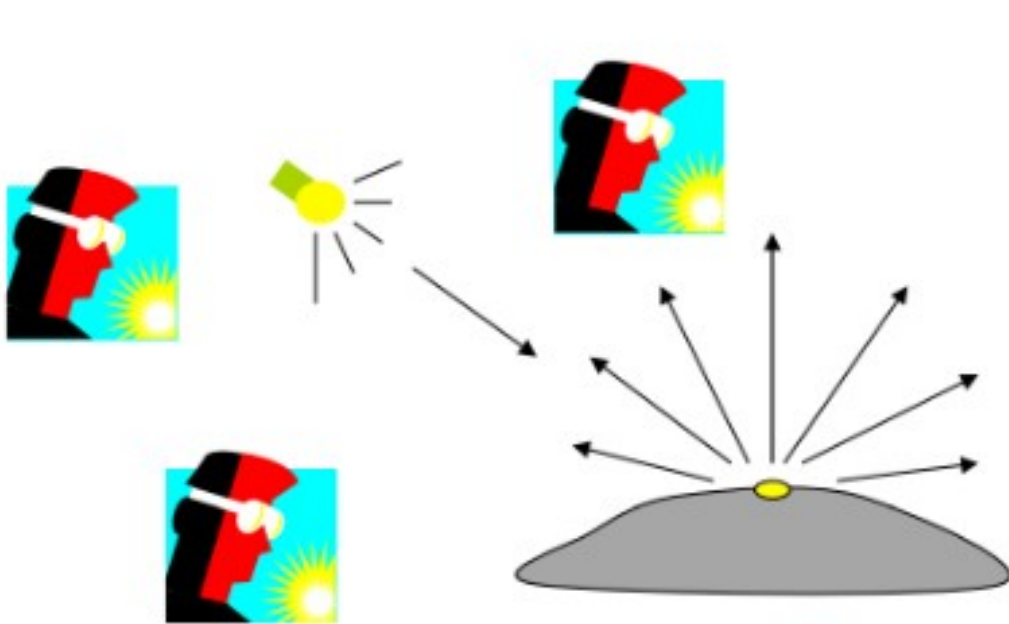


Intensity of Reflected Light

- Intensity of reflected light from a point is sum of reflected intensities
- Diffuse Reflection
- Specular Reflection
- Ambient light

Diffuse Reflection

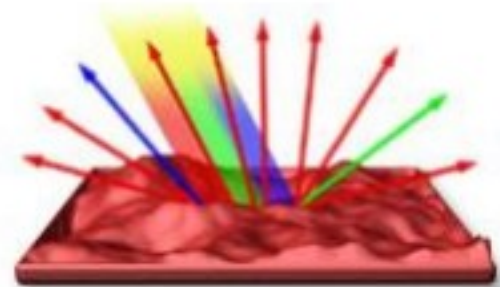
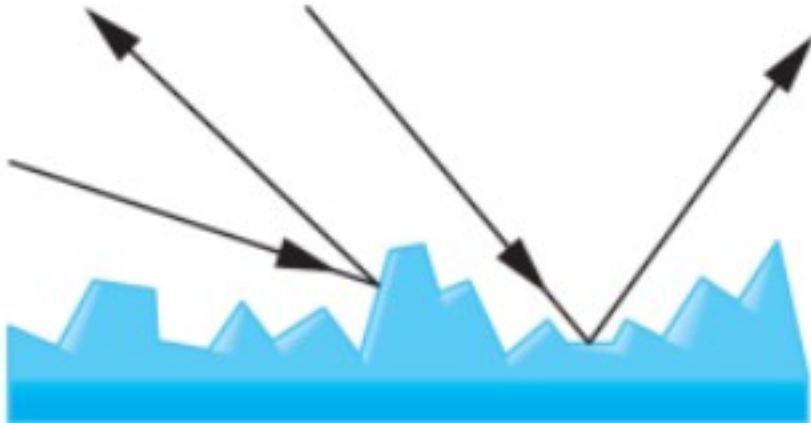
- Dull surface or grainy or rough surfaces such as soil, clay, fabric reflects the light equally in all direction as shown below.
- Viewer can see the object from any direction.



It does not matter where the eye is

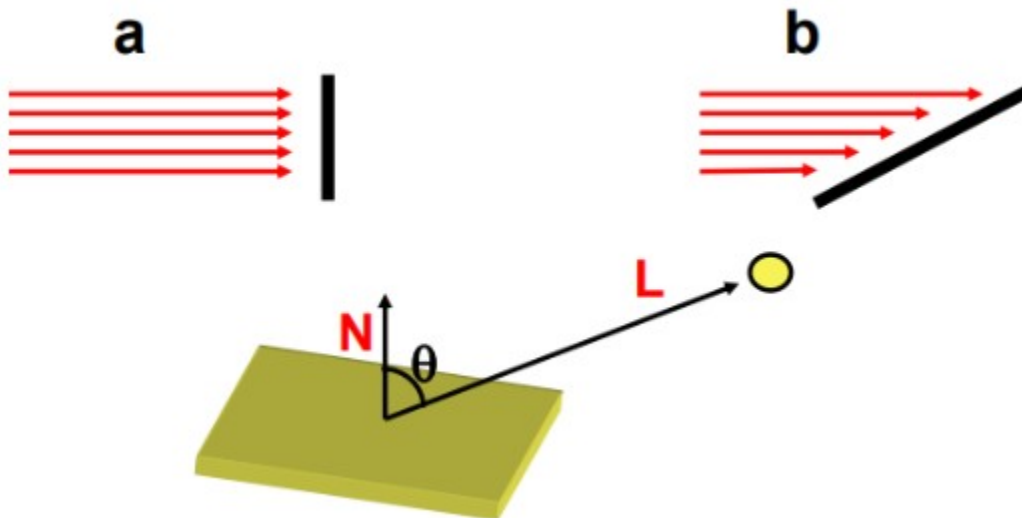
Diffuse Reflection

- Dull surfaces such as solid matte plastic reflects incoming light **uniformly in all directions**
- Called *diffuse* or *Lambertian* reflection



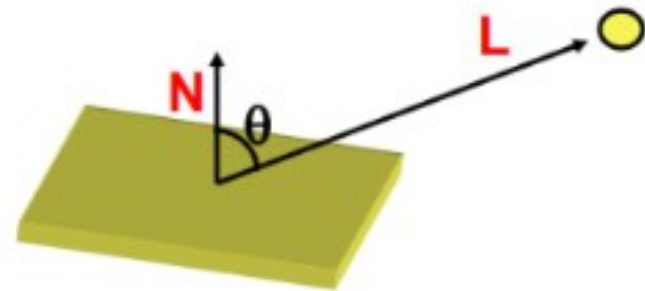
Diffuse Reflection

- So diffuse reflection follows Lambertian cosine law.
- The amount of reflected light is proportional to amount of incident light.
- a is more bright than b.
- More light get reflected at a than b.



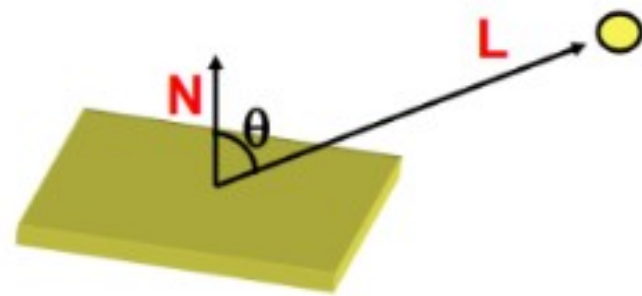
Diffuse Reflection

- Amount of incident light is $I_p \cos\theta$.
- $I_{diff} \propto I_p \cos\theta$
- $I_{diff} = k_d \cdot I_p \cos\theta$
- where,
- I_p is point light intensity
- k_d is the surface diffuse reflectivity
- k_d is based upon properties of the material of surface
- 0 means dull surface or no reflection (black color).



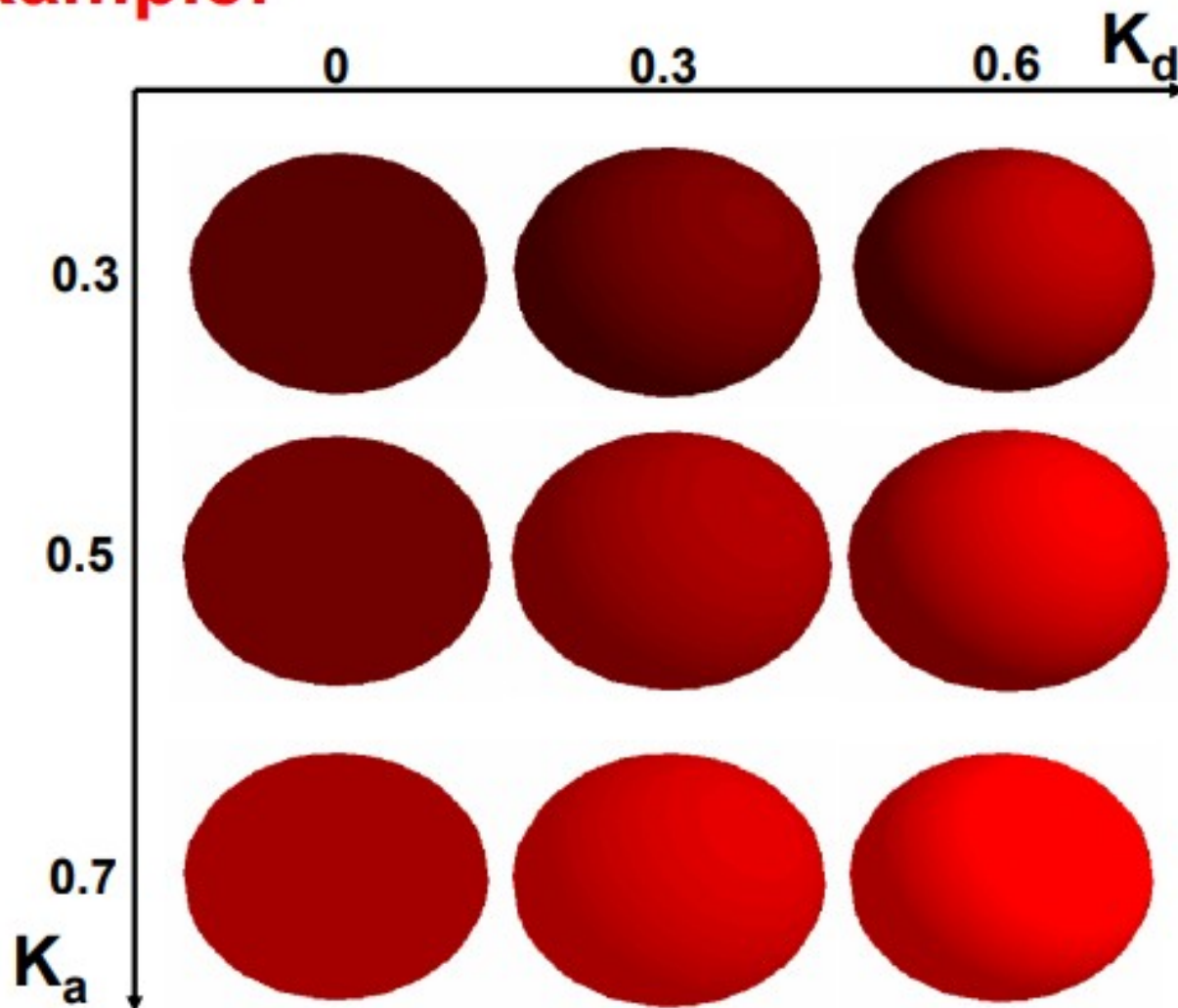
Diffuse Reflection

- 1 means shiny surface or high reflection (white color)
- N is normal vector.
- L is light vector.
- θ is angle between N and L .
- Also, $\cos\theta = \hat{N} \cdot \hat{L}$
- $I_{diff} = k_d \cdot I_p \hat{N} \cdot \hat{L}$



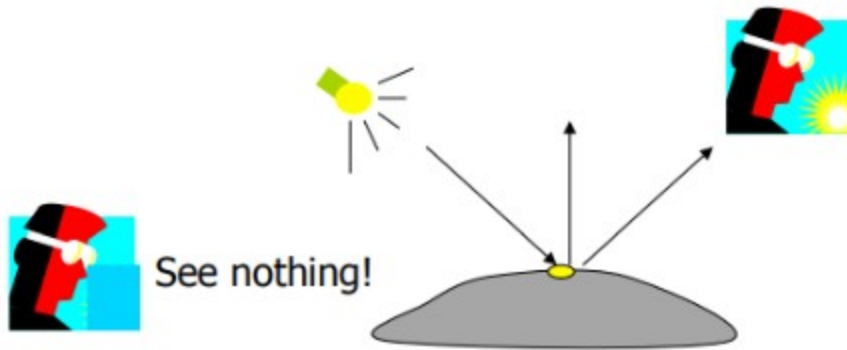
Diffuse Reflection

- **Example:**



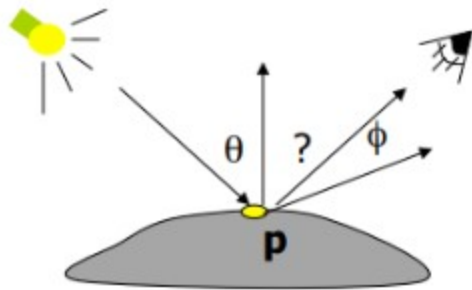
Specular Reflection Intensity

- For shiny and smooth surface, light gets reflected in only one direction.
- If we consider specular reflection only, then the viewer on the direction of reflection, It sees the object shining.



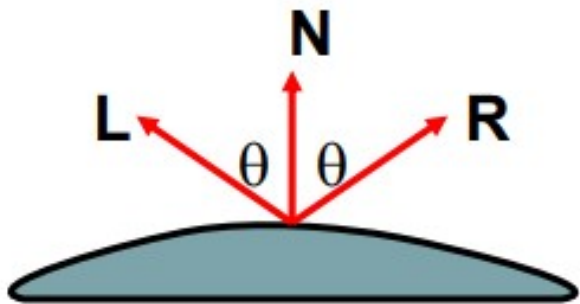
Specular Reflection Intensity

- If the viewer is little away from the specular reflection direction, the viewer sees less shininess.
- Amount of light viewer receive will be equal to incident light if viewer is at the direction of specular direction.
- So amount of incident light is

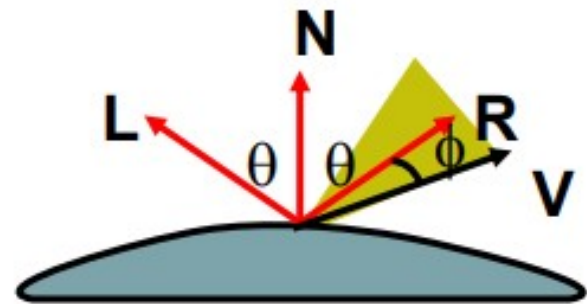


Specular Reflection Model

- An Ideal specular surface (mirror) reflects light exclusively in one direction R .
- Glossy objects (metal, plastic) reflects in the immediate vicinity of R as shown below.



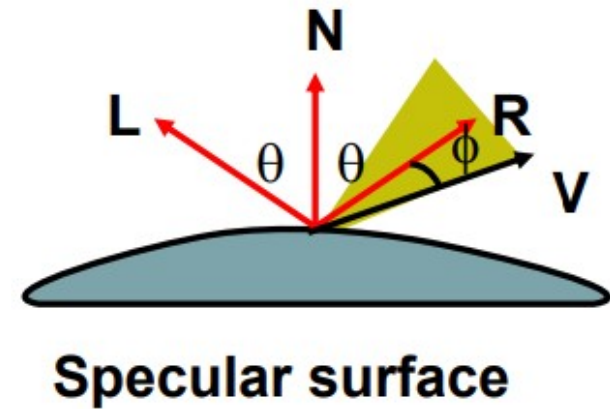
Ideal specular surface



Non-ideal specular surface

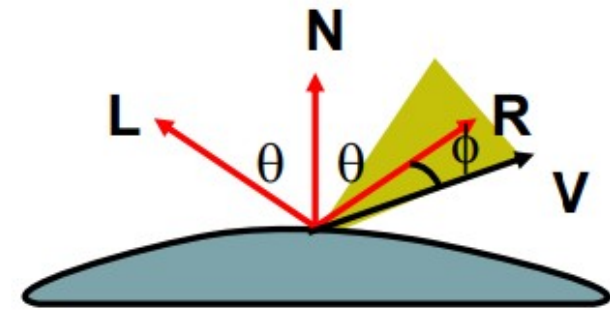
Specular Reflection

- Amount of incident light is $I_p \cos \phi$
- where, ϕ is angle between R and V
- R is specular reflection direction.
- V is view direction.
- According to Lambertian Cosine Law,
- $I_{spec} \propto I_p \cos^n \phi$
- $I_{spec} = k_s \cdot I_p \cos^n \phi$



Specular Reflection

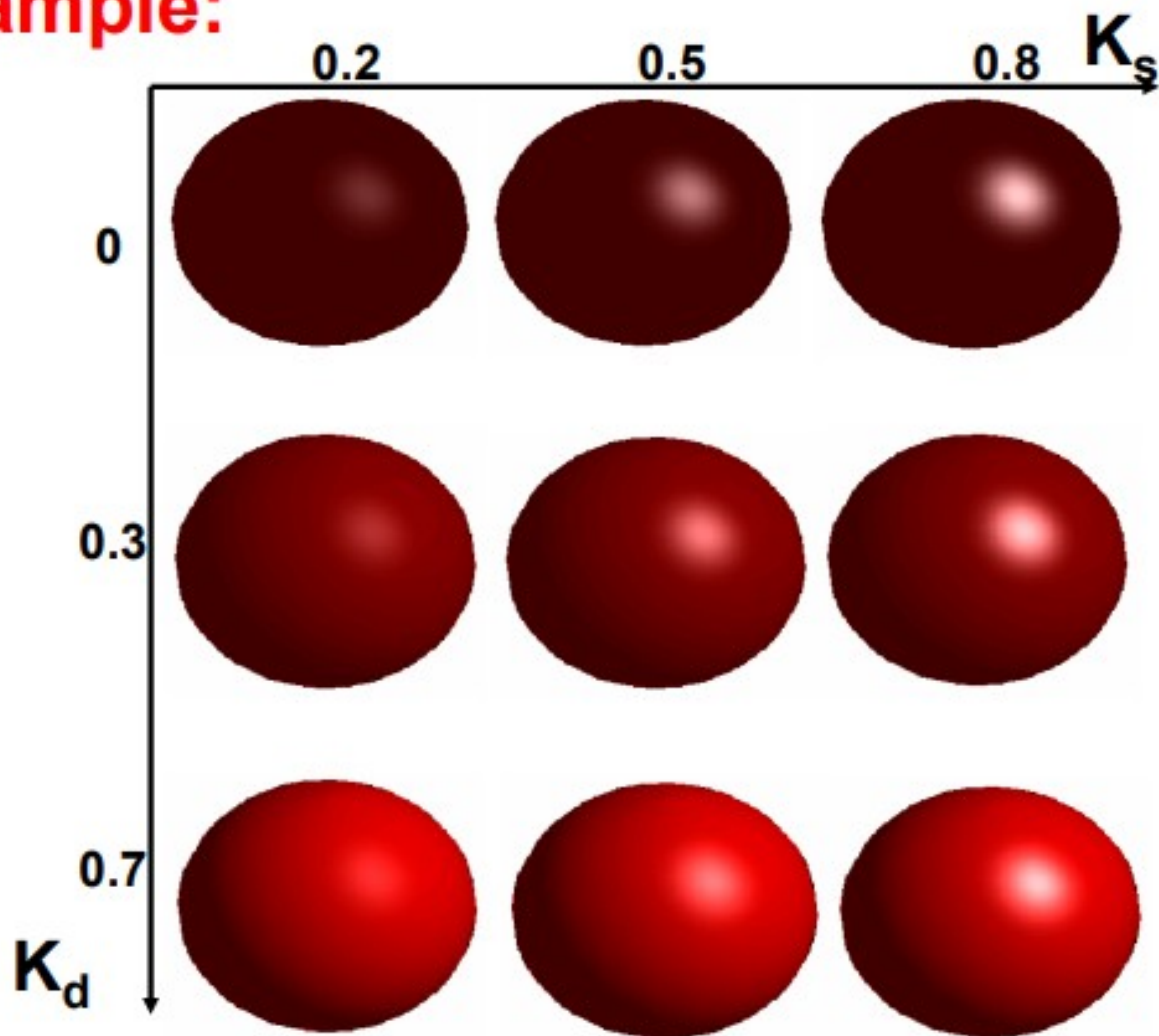
- where, I_p is point light intensity.
- n is specular reflection parameter.
- Allows to create different shining effects.
- Let $\hat{V}$ be unit vector of $\vec{V}$ and
- Let $\hat{R}$ be unit vector of $\vec{R}$
- $\cos\phi = \hat{V} \cdot \hat{R}$
- So, $I_{spec} = k_s \cdot I_p \cos^n\phi = k_s \cdot I_p \cdot (\hat{V} \cdot \hat{R})^n$



Specular surface

Specular Reflection

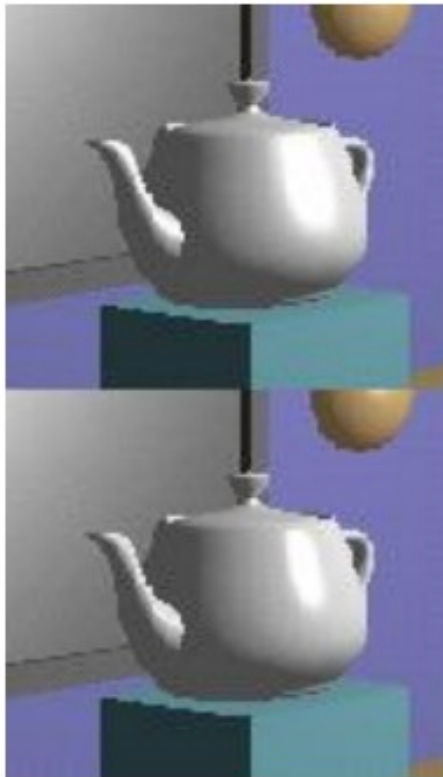
- **Example:**



Specular light calculation (3)

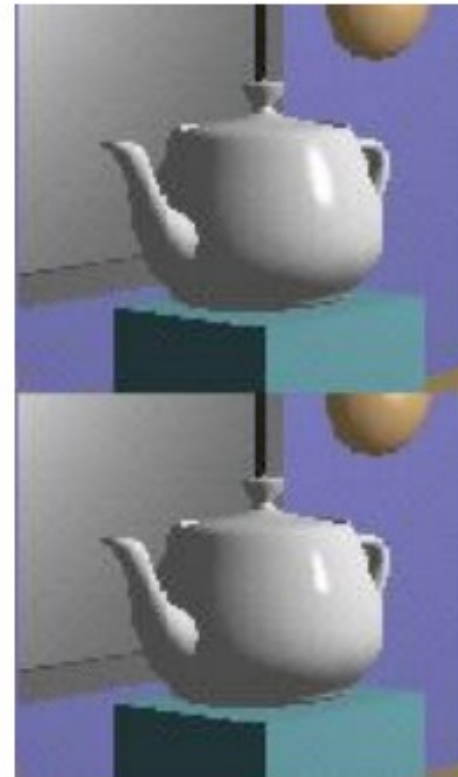
- The effect of 'n' in the phong model

n = 10



n = 30

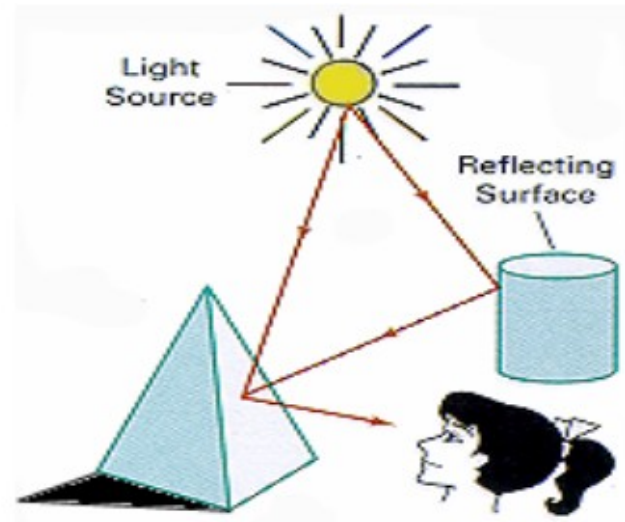
n = 90



n = 270

Ambient Reflection

- Even if an object is not illuminated with direct light source, we are still able to see it.
- The light reflected by surrounding objects illuminates indirectly.
- Such source is called ambient source.
- Independent of light position, orientation, observer's position.
- It has no direction
- It illuminates all the surface equally on the scene.



Ambient Light Model

- Different objects can reflect different amount of ambient light.
- Let us suppose the ambient light incident on a point with intensity I_a .
- The ambient reflected intensity is
- $I_{amb} = k_a \cdot I_a$
- Where, k_a is ambient reflection coefficient.
- Determines amount of ambient light reflected.
- Or, determines color.



Illumination

- Transport of light energy from source to object is called illumination.
- Illumination is used to compute color.
- Assigning color is conceptually same as illuminating scene with light.
- Intensities are real numbers.
- Incident light gets reflected and received by viewer's eye.
- Then viewer can see the object with color.

Phong Lighting Model

- Intensity of light reaching viewer from surface point is equivalent to sum of
 - Ambient reflection intensity,
 - Diffuse reflection intensity and
 - Specular reflection intensity
- $I = I_{\text{diff}} + I_{\text{amb}} + I_{\text{spec}}$
- Reflected light intensity is fraction of incident intensity.
- It is determined by surface property known as reflection coefficient or reflexivity.

Phong Lighting Model

- Coefficient controls lighting effect
- We define following coefficients
- Diffuse reflection coefficient for ambient light, k_a
- Diffuse reflection coefficient for direct light, k_d
- Specular reflection coefficient for direct light k_s
- Coefficient value varies from 0 to 1.
- 0 means dull surface where no reflection
- 1 means shiny surface where almost all incident light is reflected

Phong Illumination Model

$$I = I_{amb} + I_{diff} + I_{spec}$$

$$I = k_a \cdot I_a + k_d \cdot I_p \hat{N} \cdot \hat{L} + k_s \cdot I_p \cdot (\hat{V} \cdot \hat{R})^n$$

$$I = k_a \cdot I_a + I_p (k_d \hat{N} \cdot \hat{L} + k_s (\hat{V} \cdot \hat{R})^n)$$

For k light sources are present in scene

$$I = I_{amb} + \sum_k I_{diff}^k + I_{spec}^k$$



Intensity Attenuation

- So far we have assumed that intensity does not change as it moves from source to surface.
- If so, then intensity received by two objects, one further than another from the source, will be same.
- Color computed will also be same.
- This will result in indistinguishable distance of the object.
- It will reduce the perception of 3D also.
- So, we must consider attenuation in terms of attenuation factors



Intensity Attenuation

- Radial Attenuation factor
- Angular Attenuation factor

Radial Attenuation Factor

- It accounts for the effect of diminishing light intensity over distance.
- It is computed as given below.

$$AF_{rad} = \frac{1}{a_0 + a_1 d + a_2 d^2}$$

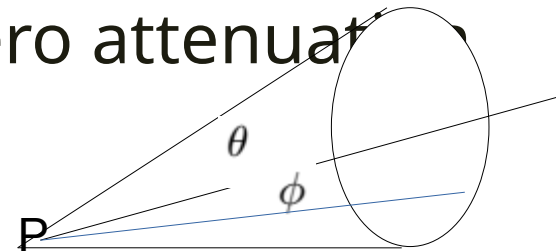
Where,

a_0 , a_1 and a_2 are coefficients that are varied to produce different effects.
 d is the distance between source and surface point.

Angular Attenuation Factor

- Used primarily to generate spotlight effect.
- It takes into account that intensity changes
- As angle between cone axis and surface line changes.
- Let a be attenuation exponent
- Let θ be angular limit
- ϕ is angle between axis and surface line of the point
- P is point source of light
- Any point outside of θ will have zero attenuation

$$AF_{ang} = \begin{cases} 0 & \text{if surface point lies outside the limit of } \theta \\ \cos^a \phi & \text{if } 0 \leq \phi \leq \theta \end{cases}$$



Modified Phong Model

- Now, Modified lighting model is
- If we set factor values to 1, then we can eliminate attenuation.
-

$$I = I_{amb} + AF_{rad} \cdot AF_{ang} [I_{diff} + I_{spec}]$$



Down sides of Lighting Model

- In graphics, generating a realistic image involves computing color at large number of points.
- This involves large computations.
- Applying lighting model will not be appropriate to generate realistic image in real time.
- This will be expensive in terms of resources such as processor and memory.
- Most graphics applications (animation, video, games) requires screen to change frequently.
- Makes the process even more complicated.
- Computation has to be done in small span of time.



Shading

- Shading models can reduce the time complexity.
- Basic idea is color of few points are computed using lighting model.
- Those colors are used to interpolate colors at other points on the surface.
- Shading models are interpolation based.
- Can be applied using efficient increment procedures.
- Lighting is applied in scene description (3D view coordinate system)
- Shading is applied in pixel level after scene is mapped to the screen.



Shading

- Flat Shading
- Gouraud Shading
- Phong Shading



Flat Shading

- Simplest shading model
- Involves least computations.
- Color of any one point is computed using lighting model.
- This color is assigned to all other points on the surface that are mapped screen pixels.
- Results in unrealistic scene.
- Not good to color any surface

Flat Shading

- Gives reasonable result only if all of the following assumptions are valid:
 - The object is a polyhedron
 - Light source is far away from the surface so that $N \cdot L$ is constant over each polygon
 - Viewing position is far away from the surface so that $V \cdot R$ is constant over each polygon





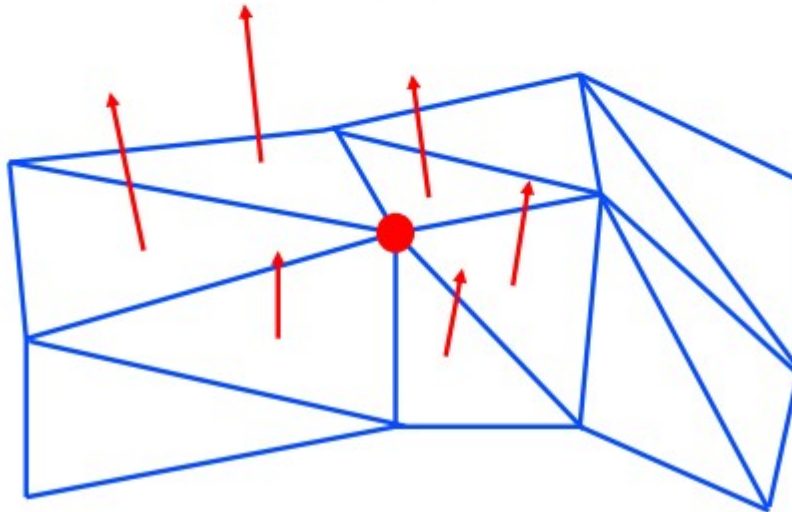
Gouraud Shading

- More realistic coloring than flat shading at the expense of increased computation.
- Determine average unit normal vector at each vertex of polygonal surface.
- Apply lighting model at each vertex to compute color.
- Linearly interpolate vertex intensities over the projected area of polygon

Gouraud Shading

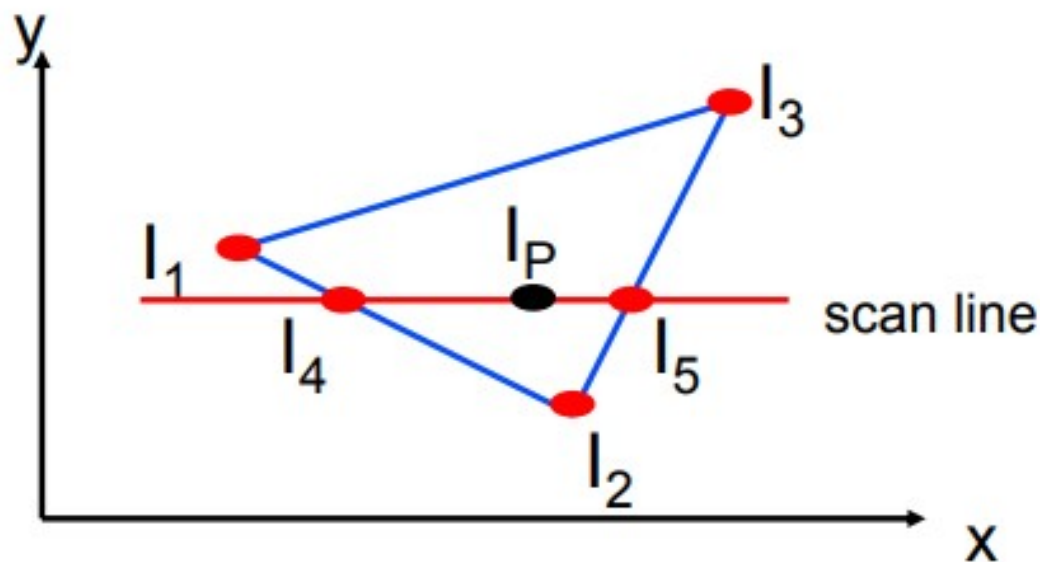
- The normal N_v of a vertex is an average of all neighboring normals:

$$N_v = \frac{\sum_k N_k}{\left| \sum_k N_k \right|}$$



Gouraud Shading

- Interpolation of the vertex intensities



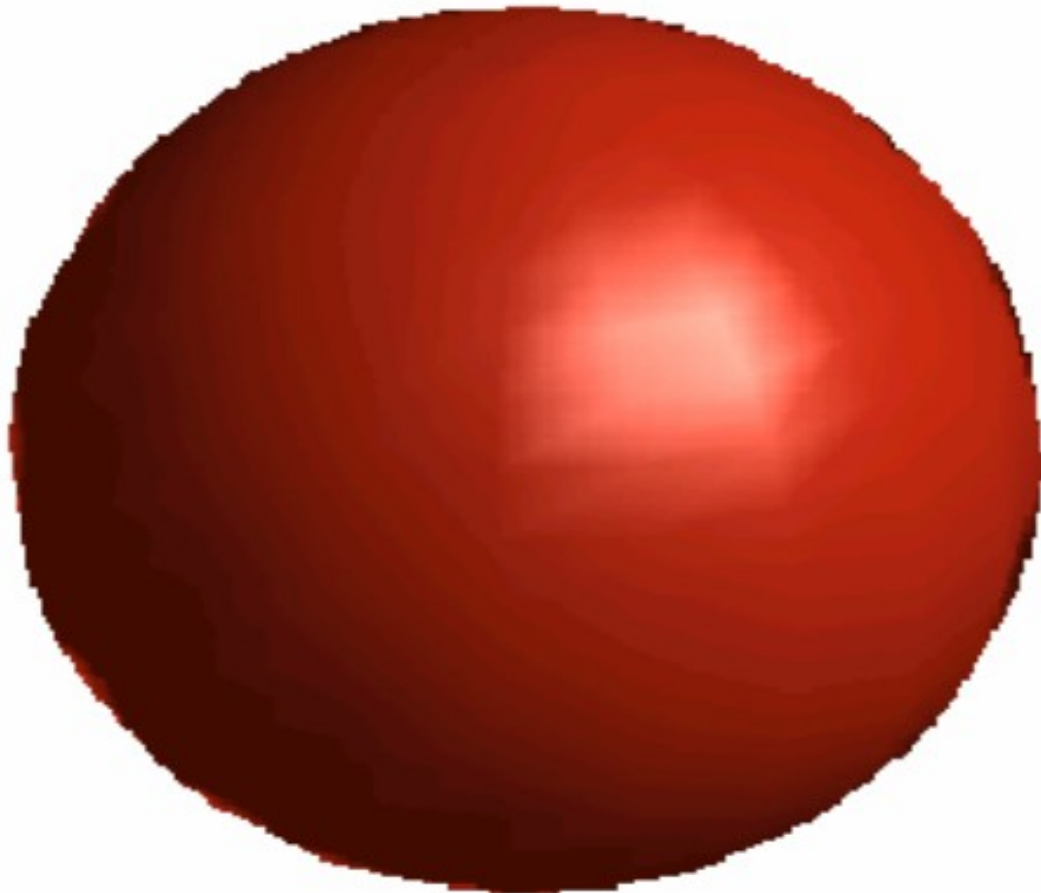
$$I_4 = \frac{y_4 - y_2}{y_1 - y_2} I_1 + \frac{y_1 - y_4}{y_1 - y_2} I_2$$

$$I_5 = \frac{y_5 - y_2}{y_3 - y_2} I_3 + \frac{y_3 - y_5}{y_3 - y_2} I_2$$

$$I_p = \frac{x_5 - x_p}{x_5 - x_4} I_4 + \frac{x_p - x_4}{x_5 - x_4} I_5$$

Gouraud Shading

- **Example:** Gouraud shading of a sphere



Phong Shading

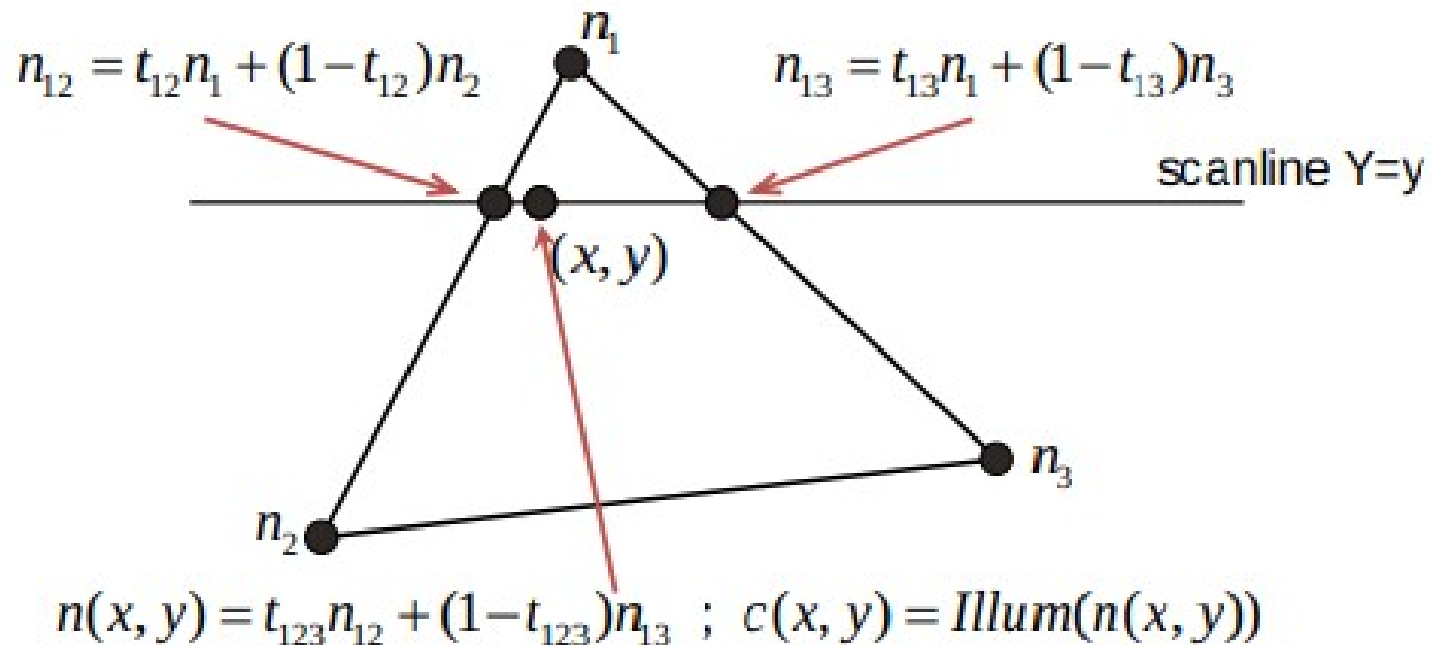
- A more accurate method for rendering a polygon surface is to interpolate normal vectors, and then apply the illumination model to each surface point

Phong Shading Algorithm:

1. Determine the normal at each polygon vertex
2. Linearly interpolate the vertex normals over the surface polygon
3. Apply the illumination model along each scan line to calculate intensity of each surface point

Phong Shading

- Interpolate (at the vertices in image space) normal vectors instead of illumination intensities
- Apply the illumination equation for each interior pixel with its own (interpolated) normal



Phong Shading

- **Example:** Phong shading of a sphere

